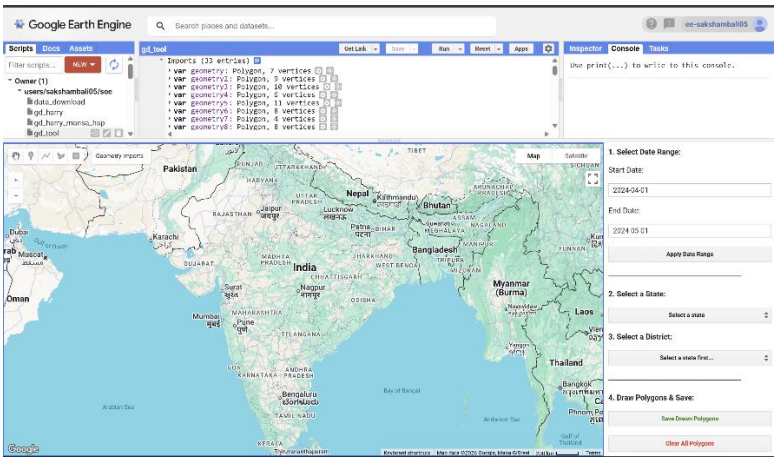
Instructions to Use the GEE Tool

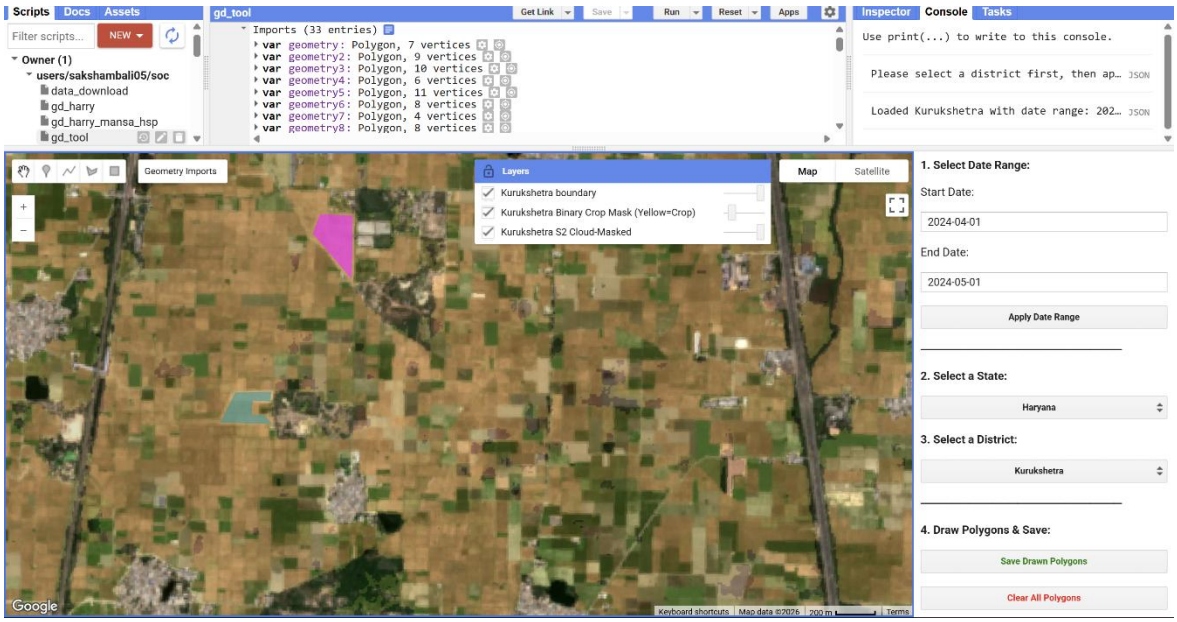
1. Setup

1. Open the Google Earth Engine Code Editor: https://code.earthengine.google.com/
2. Copy the code from the provided .js file and paste it into the editor
3. Click Run and then you will see a dashboard like this show up.



2. Workflow

4. Enter the **Start Date** and **End Date** (as per the table above, format: YYYY-MM-DD) and then click on apply date range.
5. Select the assigned **State** from the dropdown
6. Select a **District**
7. Once loaded, you will see the following:



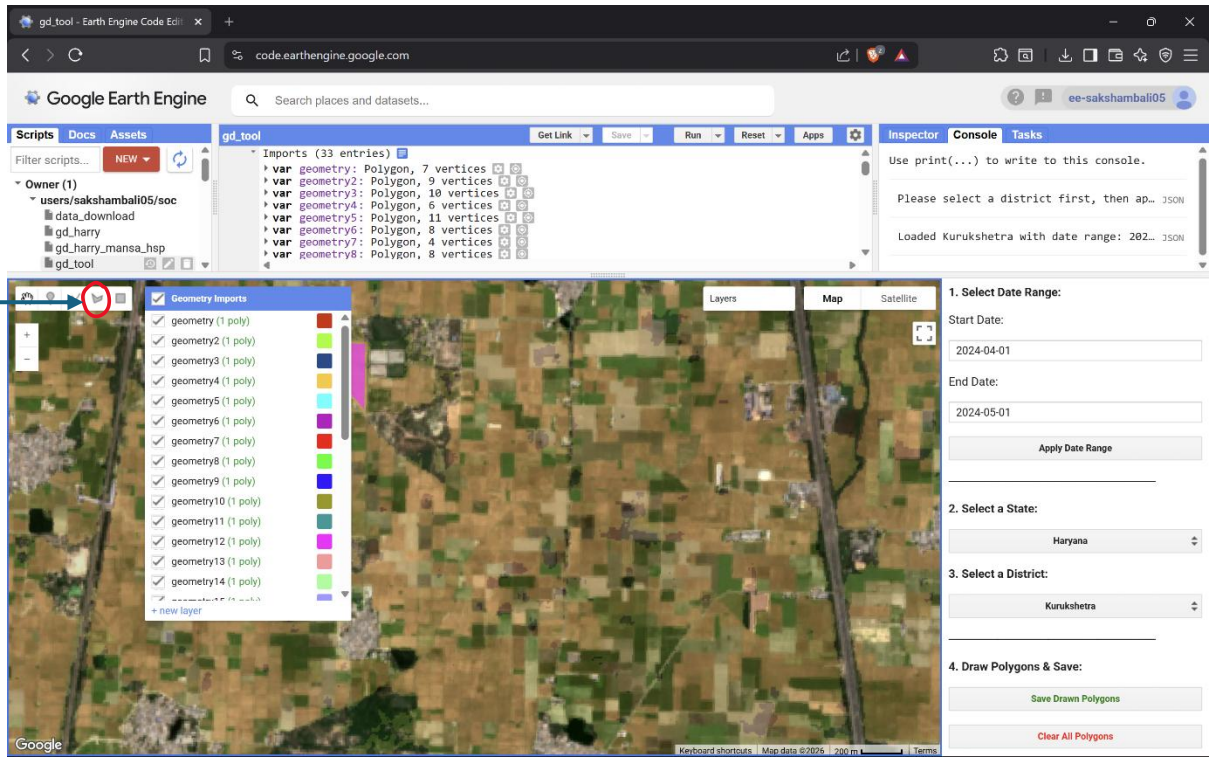
i. Layers panel (right side) (see the image above):

- Cloud-masked Sentinel-2 imagery
- Cropland mask (yellow regions indicate cropland)
- District boundary

By using the slider you can adjust the opacity of the layer

ii. Geometry Imports panel (left side):

- This will contain and store all polygons drawn during annotation (each polygon will appear as a separate layer)



3. Drawing Polygons

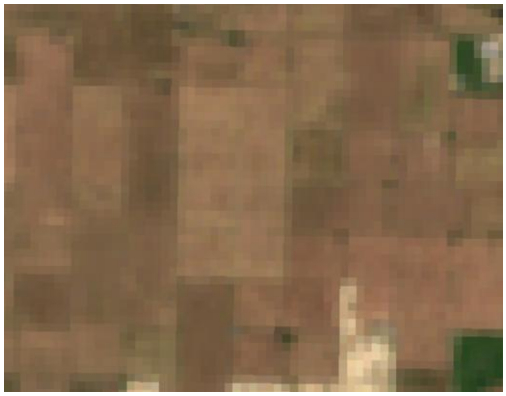
8. To draw a polygon:
   - Use the **"Draw a shape"** tool (left side near Geometry Imports)
   - Mark polygons over relevant regions

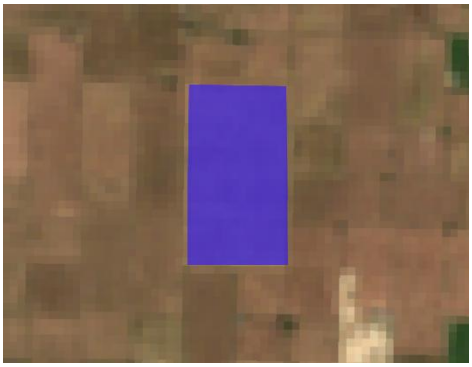
Objective:

- Select areas that are **cropland (yellow mask)**
- Ensure the Sentinel-2 image shows **bare soil (no visible crop growth)**

- Avoid selecting mixed regions (part crop, part non-crop) and prefer homogeneous patches.

Example:





9. Draw approximately **50 polygons per district**

4. Saving Data

10. Click **“Save Drawn Polygons”**
11. Go to the **Tasks** tab (top-right panel)
12. Click Run on the generated task, and confirm the export in the popup window
13. This will:

- Create a folder (if not already present): Ground_truth_SOC_<StateName>
- Save a CSV file containing polygon data for that district

14. After completing all districts for a state, please share the **Google Drive folder link** with us

Output Format

- Format: CSV
- Location: Google Drive → Ground_truth_SOC_<StateName>
- Each file contains polygon geometries for a district